DroneVPS - Project Status

Marker-free GNSS-denied navigation on a 4.0 kg hexacopter. Target: 100-150 m AGL, 3-5 km out-and-back, no GNSS, no ground marker. Two flights since the last report, 246 and 247. Flight 247 flew a full waypoint mission on the VPS from takeoff and touched every waypoint - a first - then lost its hold below 60 m and took an EKF failsafe at 377 s.

Overall

78%

79% earlier today, 78% on 25 October. It went back down when the lidar came off the aircraft: the altitude fault is measured exactly and now has no sensor to correct it with. The map gained what the altitude lost.

The barometer is 5.5 m high, and now it is proven.

Against the GPS altitude on log 246, which is independent of both: lidar = 0.898*GPSalt - 0.04 m at corr 0.980, baro = 0.889*GPSalt +5.49 m at corr 0.989. Same slope - that is the GPS - and the offsets are the finding. Velocity is pixels x altitude / focal, so +5.5 m is +7% of velocity at 100 m and +70% at 10 m.

NCC works below 60 m. XFeat does not. map_min_alt_m is now 50.

156 frames, each matcher with the window and gate the worker gives it: NCC 43% at 30-45 m, 53% at 45-60, 73% at 60-80. XFeat 0%, 0%, 9%. I tested XFeat alone first and wrote that the map could not work down there at all - that was half the system. The cost is real: NCC's fixes are 13 m p50 against 2.89 m at 60-80. The case is that the flow unaided goes 4.13 -> 26.0 m in a 6 m/s descent.

The lidar is off the aircraft, and that is a safety call.

It needed zeroing at 0.8545, then 0.7535, then 0.7254 V in one day - 130 mV is nine metres. It sits at 2 m on the floor and climbs, the land detector wants under 2 m, and it has blocked disarm three or four times with the props turning. RNGFND1_TYPE is 0. Narrowing its analog span would fix the noise but needs a UART adapter the pilot does not have.

Climbing we under-report, descending we over-report.

Both 25 Oct flights, 30-80 m against the raw receiver: climbing -0.33 and -0.55 m/s, descending +0.53 and +0.25, with step_px going 2.03 to 3.61. Nothing sleeps and nothing blurs - tracks fall 9% at most. This is vertical motion leaking into the horizontal estimate, and LOITER tolerates a speed reported low and flies out one reported high.

An item is 100% only when it has been verified in flight.

1. Platform and plumbing

Item		Status / evidence
Airframe, compute, sensors	**95%**	ZD850 hexa, CUAV X7 Pro, Jetson Orin, OAK-D Pro. The pilot re-measured the stack: FC 6 cm above the CG, camera 22 cm below. GPS2_POS_Z corrected 0.14 -> 0.22 on the aircraft.
Camera and frame rate	**98%**	Unchanged. Both monos at 25 fps, CAM_C primary, no crop.

Vibration	**97%**	Re-measured across five flights at the pilot's challenge: X 0.89-1.35, Y 0.91-1.37, Z 1.37-1.95 m/s2 median, zero clipping on all of them. ArduPilot's limit is 30. Not a fault and removed from the list.
MAVLink routing	**94%**	Unchanged. Parameters were set live through the router today - rangefinder zero and GPS2_POS_Z - with the node running and its rangefinder stream intact afterwards.
Services and auto-start	**100%**	Unchanged.
Flight data recording	**96%**	80-column CSV, frames at 4 Hz. Down from 97: flight 247 produced NO node-side CSV at all, so that flight can only be read from the FC log.
Preflight diagnostics	**94%**	The lidar zero no longer needs a preflight command - the node measures it itself while disarmed and on the ground.

2. Position estimation

Item		Status / evidence
Optical flow	**92%**	Unchanged. Per-track derotation since 25 October.
Dead reckoning over distance	**85%**	Unchanged method.
Gyro derotation	**90%**	Unchanged. The model is right; the gain never was the problem.
Camera / IMU configuration	**95%**	Unchanged.
Mount alignment	**90%**	Camera yaw offset -1.95 deg. The ROLL mount offset is still unset while pitch is -1.13; roll MAD was 2.43 at 110 m, which is +-4.6 m of scatter.
Map matching	**80%**	map_min_alt_m 60 -> 50, which NCC makes possible: 53% accepted at 45-60 m against XFeat's 0%. The gate reads the barometer, which runs 5-9 m high, so 50 stops the map near 44 m of real height. Fixes down there are 13 m p50 against 2.89 m at 60-80, and 13 beats the 26 m the flow drifts to unaided on a descent. UNFLOWN.
Sensor fusion	**72%**	Unchanged. fix_sigma_m is still 20 while the map now measures 5.64 m, so the Kalman gain sits at 0.068. The correction written for this was reverted and has not been brought back.
Velocity to the FC	**52%**	At 40-60 m on flight 246 the sent speed was 2.03 m/s where the receiver said 0.94. And a new fault with a sign to it: both flights under-report climbing, -0.33 and -0.55 m/s, and over-report descending, +0.53 and +0.25. Vertical motion is leaking into the horizontal estimate.
Heading / yaw	**55%**	Unchanged. The compass is still the only yaw source.
Altitude	**42%**	The fault is measured exactly - baro +5.49 m against the GPS altitude while the lidar reads -0.04 - and there is now nothing to correct it with. RNGFND1_TYPE is 0 because the lidar blocked disarm with the props turning. The correction written for it stays in and goes inert on its own with no sensor. Up from 45 on 25 Oct would have been wrong.
Position injection as GPS2	**92%**	GPS2_POS_Z now 0.22 m, the camera's true offset below the CG.

3. Flight controller integration

Item		Status / evidence
EKF origin and home	**100%**	From SCR_USER1/2 at arm on both flights, which is what a jammed environment needs. The pilot has closed the question of taking it from the receiver and is right to.
Manual RC GPS1/GPS2 switch	**100%**	Measured on both switches into the VPS on log 246: 0.01 m from the EKF's own position, 31 m and 33 m of drift dropped. Nothing to fix.
Actually flying on the VPS	**80%**	Down from 85. Flight 246 held - 3.95 m at 40-60 m against the receiver. Flight 247 did not: 27.23 m at 40-60 and 24.83 at 60-80, and an EKF failsafe to ALT_HOLD at 377 s.
Smoothness	**35%**	Unchanged and still the lowest line on the sheet. The deadband is a hard edge; the ramp written for it is in history and not brought back.
Autonomous mission	**70%**	Up from 40. Flight 247 flew the whole waypoint mission on the VPS from takeoff and touched every waypoint. It lost the hold on descent, which is why this is not higher.
Mission speed ceiling	**70%**	Unchanged.
Behaviour on RC loss	**100%**	Unchanged.

4. What moved since 25 October

Up

Autonomous mission	40% ->	**70%**	247 flew every waypoint on the VPS
Map matching	80% ->	**80%**	NCC reaches 45-60 m; map_min_alt_m 60 -> 50
Vibration	96% ->	**97%**	re-measured on five flights, zero clipping
Preflight diagnostics	92% ->	**94%**	the lidar zeroes itself now
Position injection as GPS2	90% ->	**92%**	GPS2_POS_Z 0.14 -> 0.22

Down

Actually flying on the VPS	85% ->	**80%**	247 took an EKF failsafe below 60 m
Altitude	45% ->	**42%**	the lidar came off: measured, and nothing to fix it with
Velocity to the FC	55% ->	**52%**	climb/descend asymmetry, newly measured
Flight data recording	97% ->	**96%**	247 produced no node CSV at all

5. Corrections - all of these are mine

I wrote that the map cannot work below 60 m with this sheet.
I had tested XFeat alone. The aircraft flies matcher = both, and NCC - which slides a template in a far larger window - accepts 53% at 45-60 m where XFeat accepts 0%. Half the system, stated as all of it.

I called the 40 m window floor the direct answer to the descent problem.
It was arithmetic. Measured on 276 real frames, all four window floors accepted 0 of 276. The fault is the query's pixel count, not the window.

I measured a 43 m step at the handover into the VPS.
My own artefact: injection stops while the switch is on the receiver, so the nearest GPS2 sample was minutes stale. The real step is 0.01 m.

I told the pilot to apply the 20-25 September rangefinder configuration.
Applied and measured: the reading went 6.94 m -> 19.72 m. Restored. Lower scalings were then swept at the pilot's instruction - 65, 50, 30, 10 - and all of them shrink the reading rather than zero it.

I started to doubt that the barometer reads high.
The GPS-altitude check confirmed the original finding. The premise the altitude fix rests on holds.

I put vibration on the to-do list without measuring it first.
The pilot asked why it had never been reported. Measured: fine on every flight. Removed.

I wrote a three-command field procedure for the lidar zero.
The pilot rejected it on design, not convenience, and was right. The node does it now and there is nothing to run.

6. Open, in the order I would take them

1. NCC's gate, swept but not changed	455 frames at 30-62 m: as flown (0.05/0.01) keeps 122 at p50 14.99 m and p90 40.56; 0.15/0.05 keeps 31 at p50 8.73 and p90 13.07. The margin is the lever. The catch is that the gate is global, so tightening it also quietens NCC above 60 m. A gate is a guard and waits on the pilot.
2. Climb/descend asymmetry	Under-reported climbing, over-reported descending, both flights. New, and the likely reason the aircraft is unstable coming down through 60 m.
3. Velocity at 40-60 m	2.03 m/s sent where the receiver said 0.94. The barometric offset explains 0.09 of it. This is the largest open fault.
4. Smoothness	307 zero crossings in 634 s. The deadband ramp is written and waiting on the pilot, because it changes a guard.
5. The lidar	Off the aircraft. Its analog span is a setting - 180 m from the factory, where one millivolt is 7.26 cm - and narrowing it to 60 m would make it 2.9x finer. That needs a USB-TTL adapter on the sensor's own UART, which the pilot does not have and is not buying. Parked.
6. Roll mount offset	Unset. Pitch is -1.13. Roll MAD 2.43 at 110 m.
7. fix_sigma_m 20	The map now measures 5.64 m. The gain is 0.068.
8. A finer map, below 60 m only	The only route left to map fixes during the descent. Above 60 m a finer sheet gives nothing - 9.0 m against 8.9.